

NVIDIA Cosmos, Explained in Depth

A beginner-to-intermediate guide to all four generations — Cosmos 1, 2, 2.5, and 3 — with diagrams, model sizes, data, inference speed, and the pretraining + post-training (SFT / GRPO) recipes.

What is a “World Foundation Model” (WFM)?

A **language** model learns from text and predicts the next *word*. A **World Foundation Model** learns from video and predicts what happens next in the *physical world* — how things move, collide, and behave. NVIDIA **Cosmos** is a family of WFMs for **Physical AI** (robots, self-driving cars): instead of testing a robot millions of times in reality, engineers let Cosmos **imagine** realistic video of what would happen and train on that.

Mental model: Cosmos is a “dream engine” for machines. Each generation dreams with higher quality, runs faster, understands your request better, and — by Cosmos 3 — even hears sound and decides physical actions.

Mini-glossary (so the rest makes sense)

Tokenizer	Compresses video into a small set of numbers (“tokens”) the model can handle, and decompresses them back to pixels.
Diffusion	Starts from random noise and repeatedly “cleans” it into a video. High quality, but many steps = slower.
Autoregressive (GPT-style)	Generates the video one chunk at a time, like typing word by word. Fast, good for real-time.
Flow-based	A cleaner, more direct cousin of diffusion: learns a straight “flow” from noise to data, often with fewer steps.
DiT	“Diffusion Transformer” — the Transformer architecture that powers the diffusion engine.
Mixture-of-Transformers	Several specialized Transformers in one model (e.g. one that <i>reasons</i> + one that <i>generates</i>).
Pretraining	Learning general knowledge from a huge, broad dataset.
SFT	Supervised Fine-Tuning: teach the model on curated question→answer examples after pretraining.
GRPO	An RL method: the model tries an answer several times; better-than-average tries get reinforced. Sharpens reasoning.

Contents

What is a “World Foundation Model” (WFM)?	1
The big picture: all four generations side by side	4
Important: Cosmos is a PLATFORM, not a single model	5
Diversity inside ONE family: Cosmos Predict 1	6
Each family evolved on its OWN cadence (not in lockstep)	7
Same family, across versions: the novelty at each step	8
How the architecture changed (diagrams)	9
Deep-dive: why did Cosmos 1 have TWO engines?	10
Keynote: the ONE factor that separates the two engines	11
Why each new generation? (the problem it solved)	12
The data: scale and diversity	13
Model sizes & key hyperparameters	14
Inputs and outputs: the “data contract” (DTO)	15
Key results: inference speed	16
Zoom-in: what is “sparse attention”? (the Cosmos 2 trick)	17
Zoom-in: what is “flow matching”? (the Cosmos 2.5 trick)	18
Deep-dive: how Cosmos 3 fuses 5 modalities (incl. audio)	19
Does Cosmos 3 reason THEN predict, or do it all at once?	20
Pretraining and post-training: SFT and GRPO	21
Key experimental results	22
Verification & source audit	23
Putting it all together	25
Appendix A — Example results (from NVIDIA's pages)	26
Appendix B — References & further reading	27

One-page cheat-sheet

The four generations (the arc)

- **Cosmos 1 — Toolbox:** two engines (diffusion + autoregressive), T5 text, separate tasks.
- **Cosmos 2 — Polish:** diffusion-only + sparse attention (up to 2.6× faster).
- **Cosmos 2.5 — Unify:** one flow model, Reason1 encoder, RL, 30s multi-camera.
- **Cosmos 3 — Leap:** two-tower mixture-of-transformers omnimodel; +audio +action.

The platform: families (not one model)

Family	Job	Technique
Predict	generate the future world	diffusion + autoregressive (1) → flow (2.5)
Transfer	controllable sim→real	diffusion DiT + ControlNet
Reason	understand & decide	autoregressive-text VLM (+ SFT/RL)
Tokenizer	compress video ↔ tokens	autoencoder (continuous + discrete)

Jargon in one line each

- **Diffusion** = sculptor: refine noise → video over many steps (top quality, slower).
- **Autoregressive** = writer: predict the next token (fast; over video tokens in Predict, over text tokens in Reason).
- **Flow matching** = a near-straight noise→data path → far fewer steps.
- **Sparse attention (NATTEN)** = each patch attends only to nearby patches → up to 2.6× faster.
- **Mixture-of-Transformers** = each modality/tower has its own weights but shares global attention.
- **SFT** = teach with curated Q→A; **GRPO** = RL that rewards better-than-average tries.

Numbers worth remembering

- **Predict 1:** diffusion 7B/14B, autoregressive 4B/12B; ~100M clips from 20M hours; ~10k H100s.
- **Predict 2.5:** 2B/14B; 200M clips; PAI-Bench 0.810; up to 30s, multi-camera.
- **Reason:** 1 = 7B/56B, 16K context; 2 = 2B/8B, 256K context, OCR + 2D/3D.
- **Cosmos 3:** Nano 16B (8B+8B), Super 64B (32B+32B); ~20T training tokens (press-reported).

One sentence: Cosmos went from “many tools that dream video” → “the best tool, polished” → “one smart unified tool” → “a reasoning, seeing, hearing, acting omnimodel.”

The big picture: all four generations side by side

	Cosmos 1	Cosmos 2	Cosmos 2.5	Cosmos 3
Released	Jan 2025	Jun 2025	Sep 2025	Jun 2026
Core engine	TWO engines: diffusion + GPT-style	Refined diffusion	ONE unified flow-based model	Mixture-of-Transformers (reason + generate)
Inputs	Text, or image/video	Text, or image/video	Text/image/video (one model)	Text, image, video, AUDIO + ACTIONS
Outputs	Future video	Video (480/704p)	Video up to 30s, multi-camera	Video + actions + audio, with reasoning
Reads text via	T5-XXL encoder	T5-style encoder	Cosmos-Reason1 (VLM)	Built-in reasoning transformer
Model sizes	Diff 7B/14B, GPT 4B/12B	0.6B / 2B / 14B	2B / 14B (+auto/robot variants)	Super / Nano / Edge
Training data	~100M clips (from 20M hrs)	+ action datasets	200M curated clips	~20 trillion tokens
Post-training	Video2World fine-tune	Action-conditioned fine-tune	Model merge + RL	SFT + RL (GRPO-style)
One-word vibe	Kitchen sink	Polish	Unify	Leap

Table 1. Each column is colour-coded and reused throughout this guide: [Cosmos 1](#), [Cosmos 2](#), [Cosmos 2.5](#), [Cosmos 3](#).

Important: Cosmos is a PLATFORM, not a single model

So far we compared “generations” (1, 2, 2.5, 3). But each generation is really a **bundle of separate model families that do different jobs**. The autoregressive 4B model, for example, is just one variant inside the **Predict** family of generation 1. There are three core “pillars” plus shared tools.

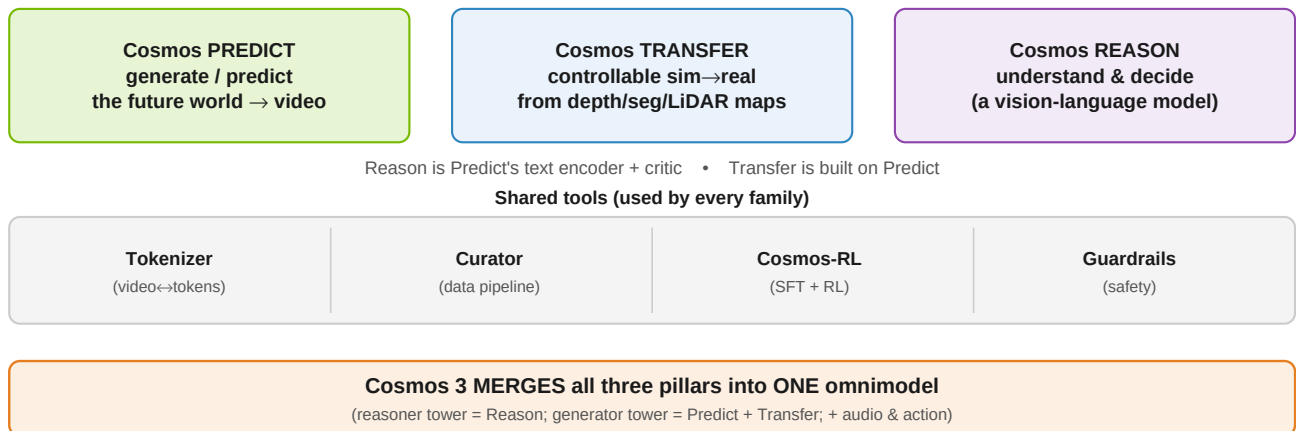


Figure 1. Three pillars (Predict / Transfer / Reason) sit on shared tools. They interconnect — and by Cosmos 3 they fuse into a single model.

The model families at a glance

Family	Its job	Input → Output	Technique	Versions
Predict	Generate / predict the future world	text/image/video → video	diffusion + AR (v1) → diffusion (v2) → flow (v2.5)	1, 2, 2.5
Transfer	Controllable generation; sim→real	structure maps (depth, seg, edge, LiDAR, HD-map) + text → photorealistic video	(Multi-)ControlNet on top of Predict	1, 2.5
Reason	Understand & decide	image/video + text → text reasoning, decisions, (v2) 2D/3D detections	vision-language model (VLM)	1, 2
Tokenizer	Compress video ↔ tokens	video → continuous/discrete tokens	causal autoencoder (FSQ)	1
Curator	Prepare training data	raw video → clean, captioned dataset	Ray GPU pipeline	—
Cosmos-RL	Train / fine-tune any family	checkpoints + data → fine-tuned models	SFT + RL	—
Guardrails	Safety	prompts/outputs → filtered content	classifiers	—

Table 2. Predict, Transfer and Reason are different MODELS for different jobs — not versions of one another. Generations 1/2/2.5 ship them separately; Cosmos 3 unifies them.

Diversity inside ONE family: Cosmos Predict 1

To show how much sits inside a single family-and-generation: **Cosmos Predict 1 alone** is about a dozen distinct checkpoints. The “autoregressive 4B” described earlier is just one of them.

Variant	Type	What it does
Diffusion Text2World 7B / 14B	diffusion	text → video
Diffusion Video2World 7B / 14B	diffusion	image/video → future video
Autoregressive 4B / 12B	GPT-style	video → next frames (the one described earlier)
Autoregressive 5B / 13B Video2World	GPT-style	text + video → future video
Tokenizer CV / DV (×3 rates each)	tokenizer	compress / decompress video
Prompt Upsampler 12B	LLM	expand short prompts into rich captions
Diffusion Decoder 7B	diffusion	clean up the autoregressive output

Table 3. One family, one generation ≈ a dozen checkpoints. Multiply by Transfer + Reason + tools to see why “Cosmos 1” is a platform, not a model.

How the families work together

- **Reason → Predict:** Reason1 is reused as the **text encoder** inside Predict 2.5, and as a **quality critic** during data curation.
- **Predict → Transfer:** Transfer 2.5 is **built on top of** Predict 2.5 — a ControlNet wrapped around it so you can steer generation with depth/segmentation/LiDAR maps.
- **Transfer's special role:** it doesn't invent freely; it **converts a structured input** (e.g. a simulator's segmentation video) into photorealistic video — the **sim-to-real** and data-augmentation workhorse.
- **Tokenizer / Curator / Cosmos-RL / Guardrails** feed and support every family.

Takeaway: don't read Cosmos as one model getting bigger. Read it as a platform of specialised families (Predict = imagine, Transfer = restyle/control, Reason = think) that generation 3 finally fuses into a single omnimodel.

Each family evolved on its OWN cadence (not in lockstep)

A subtle but important point: Predict, Transfer and Reason did **not** all version together. Predict went 1→2→2.5; Transfer jumped 1→2.5 (no public “2”); Reason has its own 1→2 track. The “generation” labels really follow **Predict**. And each generation is **not** equally “diverse”:

Era	Predict	Transfer	Reason	How diverse?
Cosmos 1 (2025 H1)	Predict 1	Transfer 1	Reason 1	Full platform
Cosmos 2 (mid 2025)	Predict 2	—	—	Mostly a Predict-only refresh
Cosmos 2.5 (Sep 2025)	Predict 2.5	Transfer 2.5	Reason 1 reused (Reason 2 later)	Full platform again
Cosmos 3 (Jun 2026)	→ generator tower	→ folded in	→ reasoner tower	Unified into ONE model

Table 4. Cosmos 2 was a Predict upgrade; 2.5 restored the full platform; Cosmos 3's novelty is that it STOPS being a platform of separate models and fuses them.

The three families converging into Cosmos 3

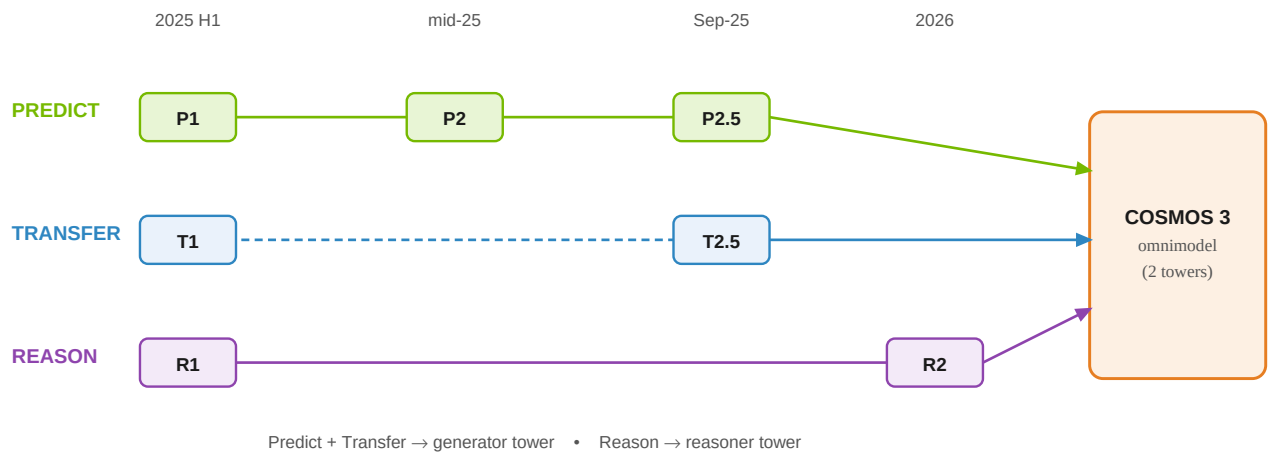


Figure 2. Dashed Transfer lane = it skipped a public “2”. All three families converge into Cosmos 3's two-tower omnimodel.

Same family, across versions: the novelty at each step

Family	Evolution (version → novelty)	Fate in Cosmos 3
Predict	1: two engines (diffusion 7/14B + autoregressive 4/12B), T5 encoder, separate tasks. 2: diffusion-only DiT (2/14B) + sparse attention → up to 2.6× faster. 2.5: flow-based + unified Text/Image/Video-to-World + Reason1 encoder + RL + 30s multiview.	Becomes the generator tower
Transfer	1: Multi-ControlNet on Predict 1; controls = depth/edge/keypoint/segmentation/LiDAR/HD-map; 4 control blocks at the START of the network; 7B. 2.5: built on Predict 2.5; control blocks DISTRIBUTED (one every 7 blocks); 2B is 3.5× smaller than Transfer1-7B with better alignment & less hallucination.	Folded into the generator tower (conditioning / control)
Reason	1: physical-AI reasoning VLM (7B/56B), hybrid Mamba-MLP-Transformer, 16K context, SFT + RL (GRPO), long chain-of-thought. 2: 256K context (16× longer), adds OCR + 2D/3D point localization + bounding boxes + stronger spatial grounding; #1 open model.	Becomes the reasoner tower
Tokenizer	1: causal autoencoder; continuous (CV) for diffusion + discrete (DV, FSQ 64k vocab) for autoregressive; up to 12× faster than prior tokenizers.	Reused as the visual/audio encoder family (ViT/VAE)

Table 5. Read each row left→right to see one family's novelty trajectory, and the right column for where it ends up inside Cosmos 3.

The one-paragraph synthesis

Cosmos 1 and 2.5 are **full platforms** (Predict + Transfer + Reason + Tokenizer); Cosmos 2 was mainly a **Predict** efficiency/quality upgrade (sparse attention); Reason ran its own **1→2** track (long context + spatial grounding). The big architectural story of **Cosmos 3** is not a new family — it is the **disappearance** of separate families: the generator tower absorbs Predict + Transfer, the reasoner tower absorbs Reason, and audio + action are added on top.

How the architecture changed (diagrams)

Cosmos 1 — two parallel engines

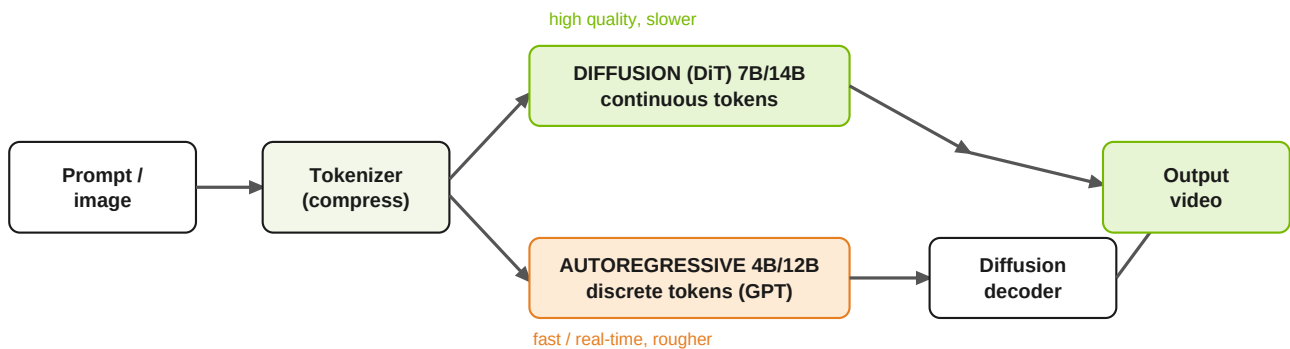


Figure 3. Cosmos 1 shipped two separate engines. Diffusion (top) = best quality; autoregressive (bottom) = fast, but needs a diffusion decoder to clean up its output. Text enters both via a T5-XXL encoder (not shown).

Cosmos 2.5 — one unified flow model with a “smart” text encoder

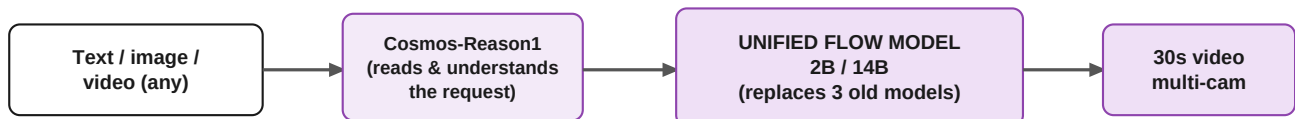


Figure 4. Cosmos 2.5 merged the three separate task-models of earlier versions into a single flow-based model, and upgraded the text encoder to Cosmos-Reason1 — a model that actually understands physics and language.

Cosmos 3 — mixture-of-transformers (thinks before it dreams)

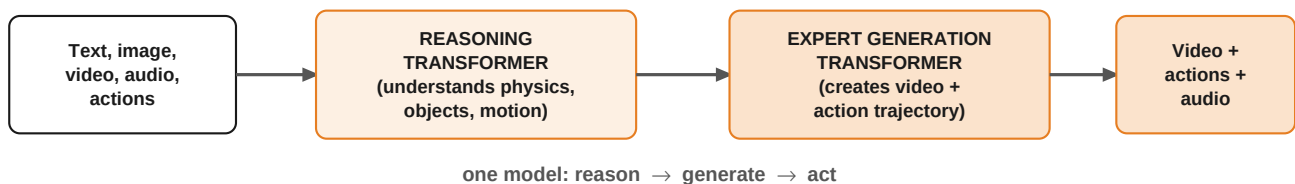


Figure 5. Cosmos 3 is no longer just a video predictor. A reasoning transformer first understands the scene, then an expert generation transformer produces video AND physical actions (and audio) — all in one “omnimodel”.

Deep-dive: why did Cosmos 1 have TWO engines?

This is the most confusing part of Cosmos 1, so it deserves a clear answer. The two engines were **not** a duplicate or a mistake — the paper states it built them on purpose: “We explore two scalable approaches for building pre-trained world foundation models — the diffusion model and the autoregressive model.” Both solve the **same** hard problem (predict believable future video) with the same trick — break one hard generation into many easy sub-steps — but in two different “spaces”, which forces two different training recipes.

*Analogy: same destination, two routes. Diffusion is a **sculptor** — start from a rough noisy block and refine the whole video over many passes. Autoregressive is a **writer** — produce the video one “token” at a time, left to right, like an LLM.*

Aspect	Diffusion engine (sculptor)	Autoregressive engine (writer)
Token space	Continuous latents (CV 8×8×8 tokenizer)	Discrete codes (DV 8×16×16; 64,000-word visual vocabulary)
Learning signal	Denoising score-matching (EDM): recover the clean signal from a noised one	Next-token cross-entropy: predict the next code (exactly like an LLM)
Text conditioning	T5-XXL via cross-attention	T5 via cross-attention added to blocks
Training stages	Text2World pretrain → Video2World fine-tune (add observed frames)	Video-only next-token pretrain → add text for Video2World
Model sizes	7B / 14B	4B / 12B (5B / 13B for Video2World)
Strength	Highest visual fidelity	Speed + streaming; plugs into the LLM toolbox (KV-cache, real-time)
Weakness	Many denoising steps → slower	Heavy compression → artifacts → needs a diffusion decoder to repaint detail

Table 6. The two engines use different tokenizers, different losses, and different training stages — they are genuinely different models, not one model trained twice.

So why keep both?

They are **complementary bets**. Diffusion is the **quality** champion; autoregressive is the **speed + LLM-integration** champion (it treats video like language — exactly the direction Cosmos 3 later embraced). The **diffusion decoder** exists only because the autoregressive path's aggressive discrete compression “can sometimes lead to undesired distortions” — so a small diffusion model cleans its output back up.

Important: does this two-engine split apply to the OTHER families?

No. The diffusion-vs-autoregressive choice is a **Predict** thing. Transfer, Reason and the Tokenizer each use the one technique that fits their job:

Family (gen 1)	Diffusion?	Autoregressive?	Core technique
Predict 1	Yes (7B/14B)	Yes (4B/12B)	BOTH — the only two-engine family
Transfer 1	Yes	No	diffusion DiT + ControlNet (wraps Predict's diffusion)
Reason 1	No	Yes — over TEXT tokens	decoder-only VLM (+ SFT/RL); “distinct from diffusion”
Tokenizer 1	No	No	autoencoder — outputs continuous (→diffusion) & discrete (→AR) tokens

Table 7. Only Predict has the two-engine split. Note “autoregressive” means different things: Predict predicts the next discrete VIDEO token (future frames); Reason is a normal LLM/VLM predicting the next TEXT token. In Cosmos 3 these become the two towers — generator (diffusion, Predict+Transfer) and reasoner (autoregressive-text, Reason).

Keynote: the ONE factor that separates the two engines

Both engines are really **conditional video generators** that learned spatial + temporal physics from video (the “world model” label is about the *goal*, not the mechanism). They differ on a **single fundamental axis** — and almost everything else follows from it: **how, and in what order, they generate.**

DIFFUSION — refine the **WHOLE** clip at once (continuous, bidirectional)



AUTOREGRESSIVE — predict the **NEXT** piece from the past (discrete, causal)

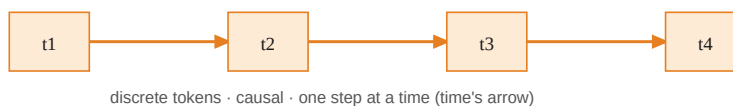


Figure 6. The deep difference: diffusion improves the entire clip together over many steps (it can see past AND future); autoregressive builds the future one step at a time from the past only — like time itself.

Everything else cascades from that one axis

	Diffusion (holistic)	Autoregressive (causal)
Representation	continuous 16-dim latents	discrete 64k codes (lossy)
Objective	denoising score-matching	next-token cross-entropy
Context	bidirectional (whole spacetime)	causal (past only)
Clip length	fixed (set up front)	arbitrary / streaming / extendable
Error over time	globally consistent, no drift	accumulates drift
Speed / quality	many steps → slower, top fidelity	KV-cache → real-time; lossy (needs decoder)
Lineage	vision / graphics	language models

Table 8. Each row is a downstream consequence of the single “generation order + representation” difference at the top.

Why it matters for a WORLD model

The physical world is **causal** — the future follows from the past, and a robot acts **step by step**. Autoregressive next-token prediction **mirrors time's arrow**, which is why it is the natural path to **interactive, streaming, real-time** world simulation. Diffusion treats a clip as a block to be perfected holistically, which is why it wins on **per-clip realism and global coherence**. That single trade-off is exactly why Cosmos kept both — and why **Cosmos 3 uses each for what it is best at**: the autoregressive **reasoner** tower (causal understanding) + the diffusion **generator** tower (high-fidelity creation).

Why each new generation? (the problem it solved)

A fair question: was each new version just a **bigger model on more data**? **No**. Scale grew every time, but it was the *enabler*, not the headline — each step introduced a distinct architectural or algorithmic idea to fix a concrete problem.

Step	Key problem with the previous version	How they solved it	Real novelty (beyond size/data)
1 → 2	Too slow, too many separate engines, visible hallucinations — a research platform, not a daily tool.	Commit to the diffusion path; add sparse attention; engineer better quality/control; add small fast variants; add action-conditioned post-training.	Sparse attention ($\approx 2.6\times$ faster) + action conditioning.
2 → 2.5	Three separate task-models to juggle; a T5 text encoder that doesn't "understand" physics; short, single-camera video.	Unify the 3 tasks into ONE flow-based model; replace T5 with the Cosmos-Reason1 VLM as encoder; add RL post-training + model merging; extend to 30s, multi-camera.	Task unification + flow matching + reasoning-VLM encoder + RL.
2.5 → 3	Still only a video predictor — can't natively reason, hear, or act; understanding and generation were separate models.	One mixture-of-transformers omnimodel: a reasoning transformer + an expert generation transformer; reasons before it generates; adds audio + action.	New architecture (MoT) + new modalities (audio/action) + merging understand/generate/act.

Table 9. Read the rightmost column: every jump is a genuine idea, not a scale-up. 1→2 buys speed; 2→2.5 buys understanding + simplicity; 2.5→3 buys a whole new capability.

The one-sentence “why”

- **1→2:** “Make it fast and reliable enough to actually use.”
- **2→2.5:** “Make it one model that truly understands the request.”
- **2.5→3:** “Stop just watching the world — reason about it, hear it, and act in it.”

The data: scale and diversity

A WFM is only as good as the video it learns from. Cosmos 1 curated **~100 million clips** out of **20 million hours** of raw video, deliberately balanced across nine kinds of physical scene so the model isn't biased toward one (e.g. only driving). Later generations grew the data by orders of magnitude.

Cosmos 1 training-data mix (by category)

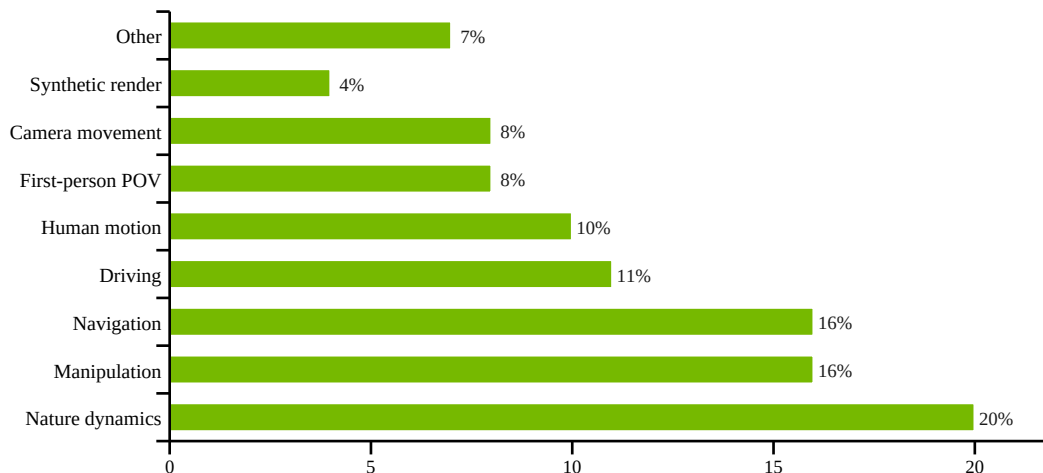


Figure 7. Diversity is intentional: nature, robot manipulation, and navigation dominate, but driving, human motion and synthetic data are all represented. This breadth is what lets one pretrained model be fine-tuned for many downstream robots/vehicles.

Data scale across generations

Generation	Pretraining data	Notes
Cosmos 1	~100M clips (from ~20M hours)	9 balanced categories; ~30% removed by dedup
Cosmos 2	Cosmos 1 data + action datasets	Bridge, AgiBotWorld, GR00T Dreams for post-training
Cosmos 2.5	200M curated high-quality clips	+ domain post-training data (auto, robotics)
Cosmos 3	~20 trillion tokens	~1B images, ~400M videos, + audio + action trajectories

Table 10. The jump from “100M clips” to “20T tokens” reflects Cosmos 3 also ingesting text, audio and robot/human action data — not just video. **Caveat:** Cosmos 3’s exact counts (20T tokens, ~1B images, ~400M videos) come from press coverage; NVIDIA’s official release states only “billions of samples across text, image, video, sound and action.”

The key insight: each generation adds a new TYPE of data

The data story is **qualitative**, not just “more”. Each generation introduces a new *kind* of data, and that new kind is exactly what unlocks the next capability:

- **Cosmos 1 — broad video** (9 balanced categories) → learns **general physics**.
- **Cosmos 2 — + action-labelled robot data** (Bridge, AgiBotWorld, GR00T Dreams) → learns **“this action causes that outcome.”**
- **Cosmos 2.5 — + multi-camera / domain data** (7-cam driving, 3-cam robotics) → learns **deployment realism and multi-view consistency**.
- **Cosmos 3 — + audio and action trajectories** (from humans and robots) → enables the **omnimodel** that can hear and act, not just see.

Model sizes & key hyperparameters

“Parameters” (B = billion) are the model's adjustable knobs — more usually means smarter but slower/heavier. Cosmos offers small *and* large versions so you can trade quality for speed.

Every model variant at a glance

Generation	Variants (parameters)	Helper models
Cosmos 1	Diffusion 7B & 14B; Autoregressive 4B & 12B (5B/13B for Video2World)	Tokenizer 77–105M; Prompt-up sampler 12B; Diffusion-decoder 7B
Cosmos 2	0.6B (Text2Image), 2B, 14B	Sparse-attention inference path
Cosmos 2.5	2B & 14B (each: pre-trained + post-trained); auto 7-cam & robot 3-cam variants	Cosmos-Reason1 as text encoder
Cosmos 3	Super (max accuracy), Nano (sub-second), Edge (local, soon)	Built-in reasoning transformer

Cosmos 1 diffusion hyperparameters (the most fully documented)

Setting	7B model	14B model
Transformer layers	28	36
Model (hidden) dimension	4,096	5,120
Attention heads	32	40
AdaLN-LoRA rank	256	256
Learning rate	2 ⁻¹⁵	2 ⁻¹⁶
Optimizer	AdamW (beta1=0.9, beta2=0.99), weight decay 0.1–0.2	
Warmup	2,500 iterations, linear	
Text encoder	T5-XXL, 512 tokens via cross-attention	
Resolution schedule	512p (57 frames) → 720p (121 frames)	
Context length @720p	56,320 tokens	
Diffusion formulation	EDM denoising score-matching; QK-RMSNorm for stability	
Compute	~10,000 H100 GPUs, ~3 months	

Table 11. NVIDIA published far more hyperparameters for Cosmos 1 than for the closed-er later releases; 2.5 and 3 reuse the same design DNA (3D-RoPE positions, AdaLN-LoRA, a causal tokenizer) at larger scale.

Inputs and outputs: the “data contract” (DTO)

Think of a **DTO** (data-transfer object) as the exact shape of what you hand the model and what it hands back. This shape widened a lot across generations.

Generation	INPUT (what you provide)	OUTPUT (what you get back)
Cosmos 1 (diffusion)	Text → T5-XXL embedding (512 tokens) + optional conditioning frames + fps + frame-count	Continuous latent video → decoded to RGB frames
Cosmos 1 (autoregressive)	Discrete video tokens (DV 8×16×16) + T5 text embedding	Next discrete tokens → diffusion-decoder → RGB frames
Cosmos 2	Text, or image, or video + fps	RGB video at 480p / 704p, 10–16 fps
Cosmos 2.5	Text / image / video, encoded by Cosmos-Reason1	Up to 30s video; optional multi-camera (3–7 synced views)
Cosmos 3	Text + image + video + ambient audio + action trajectories	Video + action trajectory + audio, preceded by explicit reasoning

Table 12. The input distribution broadens from “text + frames” (Cosmos 1) to a full five-modality stream including audio and actions (Cosmos 3). The output likewise grows from “a short video” to “video + the physical actions a robot should take”.

Why the input distribution matters

- **Cosmos 1–2:** trained mostly on curated *internet-style* video across 9 categories — great general physics, but you fine-tune for your specific robot.
- **Cosmos 2.5:** 200M *physical-AI-focused* clips + domain post-training (driving, manipulation) shift the distribution toward real deployment.
- **Cosmos 3:** adds *action trajectories from real humans and robots* and *audio* to the training mix — so the input/output distribution now includes the robot's own behaviour, not just what a camera sees.

Key results: inference speed

Speed is the headline practical difference. Faster generation = more simulations per day = faster robot/AV development. Each generation attacked latency differently.

Generation	Speed lever	Reported result
Cosmos 1	Causal, factorized tokenizer	Up to 12× faster encode/decode vs. prior tokenizer (CogVideoX)
Cosmos 2	Sparse attention in the DiT	Up to 2.6× end-to-end inference speedup
Cosmos 2.5	Flow-based sampling (fewer steps) + smaller 2B option	Faster, more resource-efficient than 2.x diffusion
Cosmos 3	Dedicated Nano variant	High-quality video + action reasoning in a fraction of a second

Table 13. Two concrete, comparable speedups are published: Cosmos 1's tokenizer (up to 12×) and Cosmos 2's sparse attention (up to 2.6×). 2.5 and 3 emphasise efficiency via flow sampling and a sub-second Nano tier rather than a single headline multiplier.

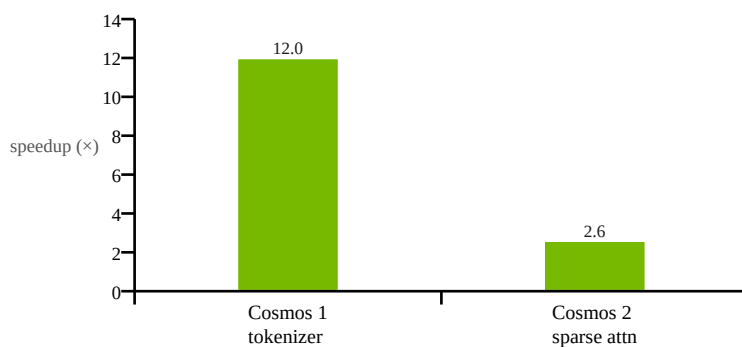


Figure 8. The two directly-quoted “× speedup” numbers. (Different things are being sped up — tokenizer vs. whole pipeline — so this compares *magnitude of improvement*, not absolute latency.)

The quality/speed trade-off in one sentence

Diffusion (Cosmos 1–2) = many denoising steps = top quality but slower; **autoregressive** (Cosmos 1) = stream tokens = fast but rougher; **flow-based** (Cosmos 2.5) = fewer steps for similar quality; **Nano** (Cosmos 3) = engineered for sub-second responses.

Zoom-in: what is “sparse attention”? (the Cosmos 2 trick)

Cosmos 2's headline speedup (up to 2.6×) comes from **sparse attention**. Here is the idea with no maths.

The problem: “attention” is very expensive for video

Inside a Transformer, every token (think: every little patch of the video) **looks at every other token** to decide what is relevant. That is called **full attention**. A short 720p video is **tens of thousands of patches** (frames × height × width), and full attention compares *every patch with every other patch* — so the cost grows **quadratically**: double the patches → **4×** the work. For video, this one step dominates the whole generation time.

The insight: most of those comparisons are wasted

A patch in the top-left of frame 1 has almost nothing to do with a patch in the bottom-right three seconds later. What actually matters to a patch is its **neighbours** — nearby in space, and a frame or two before/after.

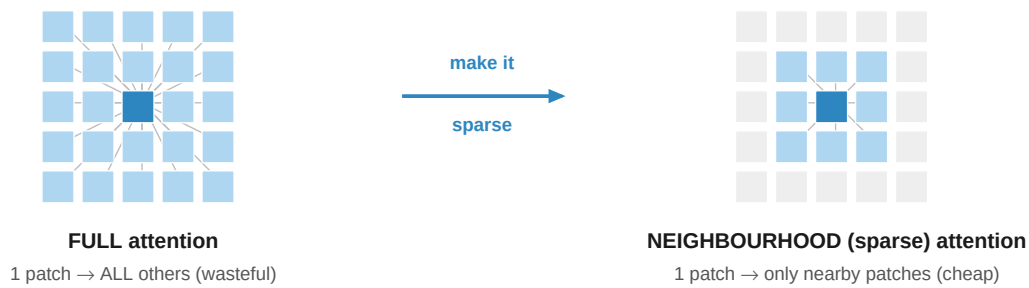


Figure 9. Left: every patch attends to all others (lines everywhere). Right: each patch attends only to its local window — the grey patches are skipped. Cosmos 2 (via the NATTEN library) drops up to 98% of the connections, keeping only the ~2% that matter.

The fix: Neighbourhood Attention (NATTEN)

Instead of attending to **all** patches, each patch attends only to those in a small **local window** around it (in space and time). Cosmos 2 pushed this so **up to 98% of the attention connections are dropped** (sparsity raised “from 50% to 98%”) — only the most relevant ones are computed.

- **Result:** far fewer calculations → **1.7×–2.6× faster** at 720p, with quality essentially preserved (the dropped links were the unimportant ones).
- **Bonus:** it is an inference-time efficiency change, not a bigger model — related “Generalized Neighbourhood Attention” can even be plugged into existing models for 28–46% speedups **without any retraining**.

Analogy: full attention = everyone in a stadium trying to talk to everyone else at once (chaos, won't scale). Neighbourhood attention = each person only talks to their own row and the rows just ahead/behind — you keep all the context that matters, and it scales.

Zoom-in: what is “flow matching”? (the Cosmos 2.5 trick)

Cosmos 2.5 switched from **diffusion** to a **flow-based** model. Here is what that means, again with no maths.

Diffusion takes a wandering path

Recall the diffusion “sculptor”: it turns noise into video with **many tiny denoising steps**. The path it follows from pure noise to a finished video is **curved and wandering**, so it needs *lots* of small steps (often 30–50+) to stay on track. Each step is a full pass through a huge model — so many steps = slow.

Flow matching learns a straight “current” instead

Picture noise on the left and real video on the right. Flow matching teaches the model a **direction to move** at every point — “which way, and how fast, to get from noise toward data”. It trains by connecting each noise sample to a real sample with a **straight line** and learning to follow it. Because the target paths are **nearly straight**, you can take **big steps** and still land in the right place — so you need **far fewer steps**.

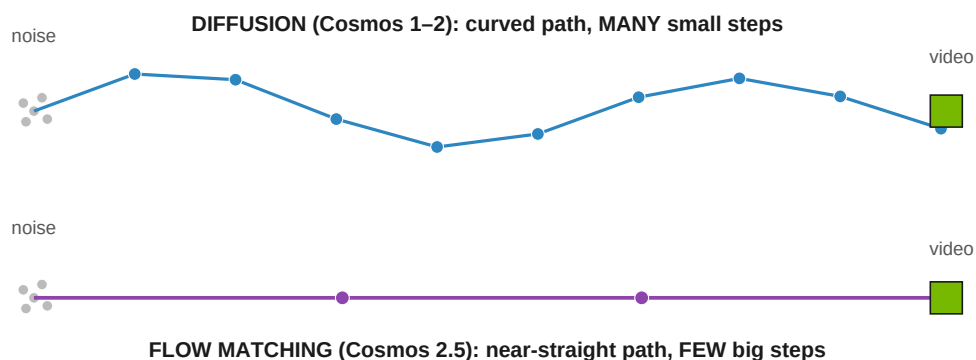


Figure 10. Same start (noise) and destination (video). Diffusion zig-zags there in many small steps; flow matching learns a near-straight route it can cover in a few big steps.

Why Cosmos 2.5 switched to it

- **Fewer sampling steps** → **faster & more resource-efficient** — important for 30-second, multi-camera video.
- **A cleaner, more stable training objective** (simply “match the direction”), which made it easier to build **one unified model** for Text/Image/Video-to-World instead of three.

Analogy: diffusion is a windy mountain road with dozens of turns (many careful steps); flow matching is a straight highway to the same town — fewer, bigger moves to arrive.

Deep-dive: how Cosmos 3 fuses 5 modalities (incl. audio)

Cosmos 3 must ingest AND produce **five** very different data types — text, image, video, **audio**, **action** — and reason about them together (text is symbols, video is pixels, audio is a waveform, action is control vectors). How do you put all of that into **one** model? Three ingredients:

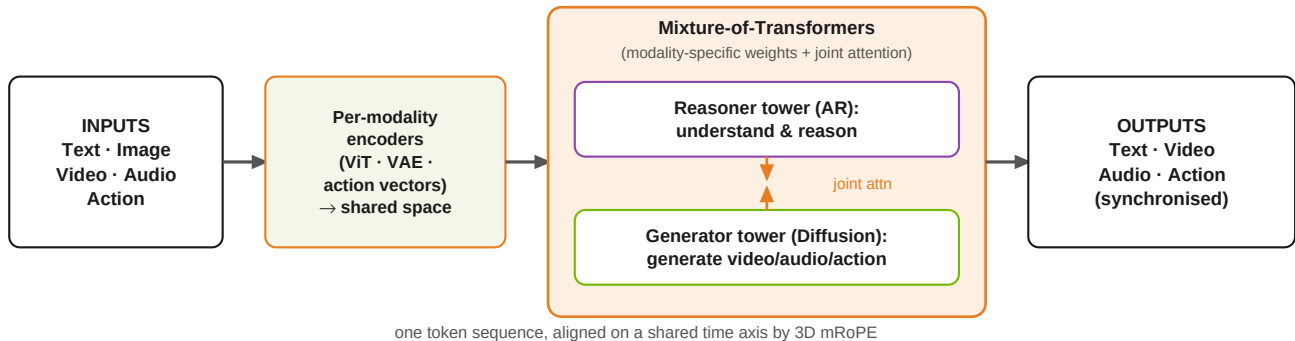


Figure 11. Each modality is encoded into a shared space, concatenated into one sequence, and processed by a two-tower Mixture-of-Transformers: separate weights per modality/tower, but one global attention so audio, video, text and action all “see” each other.

1) A dedicated encoder per modality → one shared space

Modality	How it enters the shared space
Text	Token embeddings (like an LLM)
Image / Video	A ViT for understanding; a VAE for generation
Audio	A VAE encodes the sound, then a linear projection maps audio tokens into the hidden dimension
Action	Domain-aware action vectors (robot / AV control)

2) One sequence + global attention, aligned by 3D mRoPE

All tokens are concatenated into **one sequence** with **global self-attention** — an audio token can attend to a video token can attend to an action token. A unified **3D mRoPE** puts video, audio and action tokens **on one shared temporal axis** — literally how the model knows which sound goes with which frame and which action.

3) Mixture-of-Transformers — the “one group” mechanism

In a plain transformer all modalities fight over the same weights. **MoT** gives each modality/tower its own **weights** (feed-forward, attention projections, layer-norms) **while sharing global attention** — specialisation without fragmentation. The **Reasoner** (AR = understand) and **Generator** (diffusion = create) towers use separate parameters but **joint attention**; the reasoner can run alone, generation activates both. Sizes: **Nano 16B** (8B+8B), **Super 64B** (32B+32B).

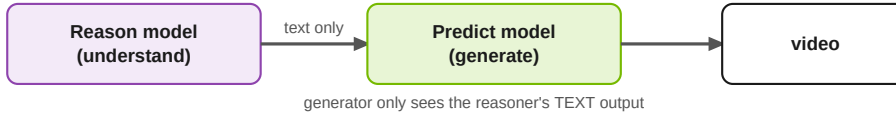
Managing the audio dataset specifically

- Audio is **ambient sound paired with video**, kept **temporally synchronised** and VAE-encoded; aligned to video/action via the shared mRoPE time axis.
- Curated through the same **multi-stage pipeline** (filter + quality review) as the rest of the data, from NVIDIA-owned + commercially-permissive sources.
- NVIDIA explicitly lists “**inaccurate sound–video alignment**” as a known failure mode — audio↔video timing is the central hard problem (and why mRoPE matters). *Granular audio-dataset counts are not publicly disclosed.*

Does Cosmos 3 reason THEN predict, or do it all at once?

A common question: are reasoning and generation two separate steps? **No — they happen jointly in one forward pass.** This is the key difference from the old platform, where Reason and Predict were literally **two separate models run in sequence.**

OLD (Cosmos 1 / 2.5): two separate models, staged



COSMOS 3: ONE model, ONE forward pass (joint attention)

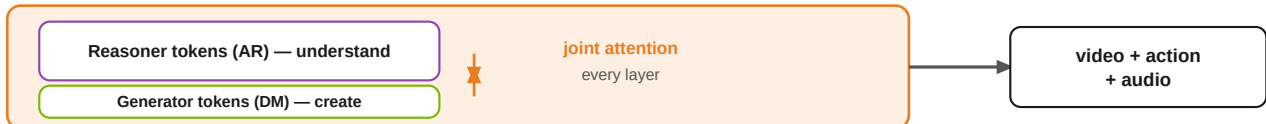


Figure 12. Old = a pipeline of two models; the generator only received the reasoner's text. Cosmos 3 = one model where reasoning and generation tokens share attention at every layer.

The precise picture

- **One unified forward pass:** NVIDIA states Cosmos 3 “can reason and generate ... in one unified forward pass” — not a two-model pipeline.
- **Separate weights, joint attention:** the two towers have their own parameters (the Mixture-of-Transformers trick) but “interact through joint attention” at every layer — so they are **coupled, not deferred** to separate stages.
- **“Reasons before generating” is an INFORMATION order, not a separate model call:** the reasoning tokens sit earlier in the sequence and the generation tokens attend back to them, so reasoning **conditions** generation within the same pass. (Typically causal attention over the reasoning tokens + bidirectional within the diffusion block — the standard pattern for hybrid AR+diffusion models; NVIDIA hasn't published the exact masking for Cosmos 3.)
- **The one staged case:** the reasoner **can run alone** as a pure VLM (understanding only). But whenever it generates, **both towers run together.**

Why joint beats staged: in the old pipeline the generator only saw the reasoner's text summary; in Cosmos 3 it can attend to the reasoner's **full internal representations**, and reasoning can be informed by what is being generated — tighter coupling means fewer hallucinations, better physical consistency, and one efficient pass instead of two model invocations.

Pretraining and post-training: SFT and GRPO

Modern Cosmos models are built in two phases. **Pretraining** soaks up broad knowledge from massive data. **Post-training** then sharpens behaviour with curated examples (**SFT**) and reinforcement learning (**GRPO**). This recipe is clearest in **Cosmos-Reason1** — the reasoning brain that powers Cosmos 2.5's text understanding and the spirit of Cosmos 3.

The 4-stage training pipeline (Cosmos-Reason1)

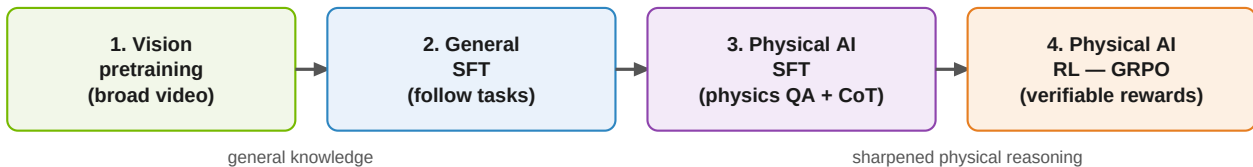


Figure 13. Knowledge first (stages 1–2), then physical-world specialisation (stage 3), then reinforcement learning to make the reasoning reliable (stage 4).

What happens in SFT (stage 3)

- Curated **question** → **answer** examples for physical common sense (space, time, basic physics) and embodied decisions (what should the robot do?).
- Built from real robotics/AV datasets: **BridgeData V2**, **RoboVQA**, **AgiBot**, **HoloAssist**, and driving data.
- The team hand-picked **~200–250 examples per data source**, turned them into multiple-choice questions, and removed ambiguous ones — quality over quantity.
- Answers include **chain-of-thought** (the model explains its reasoning step by step).

What happens in GRPO (stage 4) — in plain words

GRPO (Group Relative Policy Optimization) is reinforcement learning *without* a separate “judge” model. For each question the model produces **several attempts**; because these questions have a **verifiable correct answer**, each attempt can be auto-graded. Attempts that score **above the group average** are reinforced; below-average ones are discouraged. A **reference model** and **reward normalization** keep the updates stable so the model improves without “drifting”. The payoff: noticeably better physical reasoning.

Key experimental results

Cosmos-Reason1: bigger model & RL both help

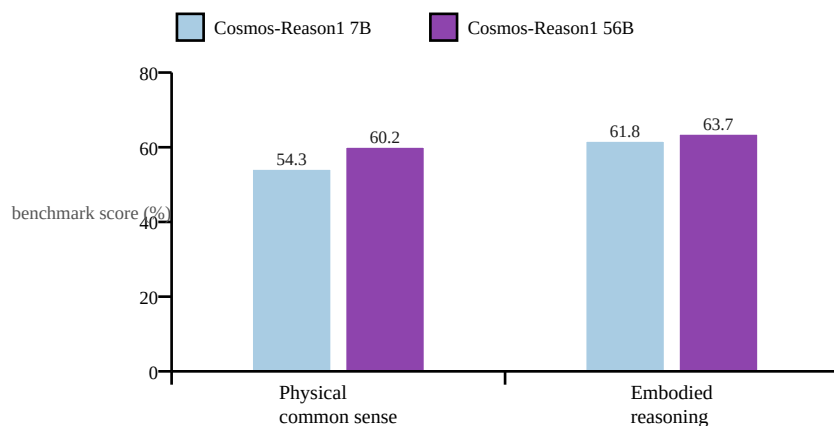


Figure 14. The 56B model beats the 7B on both physical-common-sense and embodied-reasoning benchmarks. Embodied reasoning improved by ~+10–11 points over the next-best prior model. **Source note:** numbers are from NVIDIA's Cosmos-Reason1 research page (released 7B checkpoint). An earlier arXiv v1 of the paper reported an 8B model with somewhat different scores — see the verification page.

Reinforcement learning (GRPO) lifts reasoning further

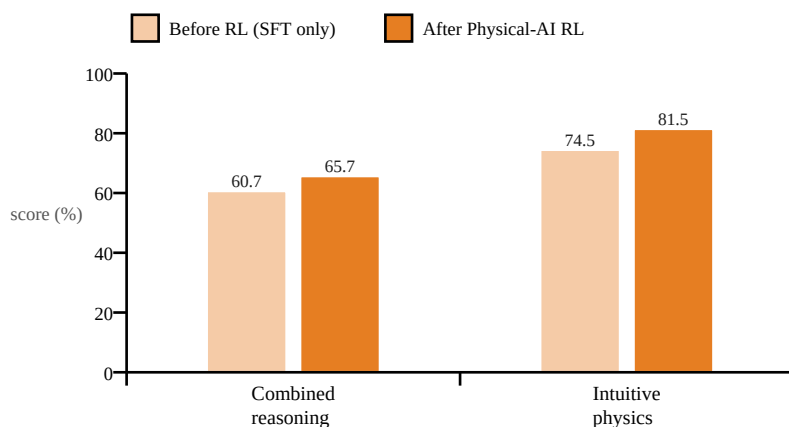


Figure 15. Adding the GRPO reinforcement-learning stage raised the 7B model by +5.0 points (combined reasoning) and +7.0 points (intuitive physics) — evidence that post-training, not just size, drives physical reasoning. **Source note:** research-page figures; arXiv v1 reports different magnitudes (e.g. intuitive physics 65.7→68.7). The *direction* — RL helps — is consistent across both.

Generation-quality results (Cosmos 2.5 & 3)

- **Cosmos 2.5** post-trained **14B** scores **0.810** on PAI-Bench Image2World (matching the strong Wan2.2-A14B baseline).
- **Cosmos 2.5** autonomous-driving multiview model: up to **2.3×** better FVD/FID (video realism metrics) versus its predecessor.
- **Cosmos 3** ranks **#1 among open models** on Artificial Analysis, Physics-IQ, PAI-Bench and R-Bench (world generation); RoboLab and RoboArena (action policy); and VANTAGE-Bench and TAR (vision understanding).

Note: FVD/FID measure how realistic/consistent generated video is (lower is better, so a 2.3× improvement is large). PAI-Bench scores world-generation quality (higher is better).

Verification & source audit

Every quantitative claim in this guide was cross-checked against primary sources (NVIDIA papers, research pages, and the official Cosmos 3 release). Of 21 checked claims, **19 verified** against a primary/official source; **2 carry caveats** (marked “!”). No factual errors were found in any claim that could be checked against a primary source.

#	Claim in this guide	Checked against	Verdict
1	Cosmos 1 data mix: 20/16/16/11/10/8/8/4/7%	arXiv 2501.03575	OK — exact
2	~100M clips (10 ⁸) from ~20M hours of video	arXiv 2501.03575	OK — exact
3	Diffusion 7B = 28L / 4096 / 32 heads; 14B = 36L / 5120 / 40	Table 11	OK — exact
4	AdaLN-LoRA rank 256; LR 2 ⁻¹⁵ / 2 ⁻¹⁶ ; context 56,320	arXiv 2501.03575	OK — exact
5	Autoregressive 4B / 12B (+ 5B / 13B Video2World)	arXiv 2501.03575	OK — exact
6	Trained on ~10,000 H100 GPUs for ~3 months	arXiv 2501.03575	OK — exact
7	Tokenizer up to 12× faster; sparse attn (NATTEN) 50→98% sparsity, 1.7–2.6×	arXiv 2501.03575; arXiv 2504.16922	OK
8	Cosmos 2: 0.6B/2B/14B; 480/704p; flow matching in 2.5	GitHub + NVIDIA blog	OK
9	Cosmos 2.5: flow; Reason1 encoder; 2B/14B; 200M clips; 30s; PAI-Bench 0.810; 2.3× FVD/FID	NVIDIA research page	OK
10	GRPO post-training; 4-stage pipeline; verifiable rewards	arXiv 2503.15558	OK
11	Cosmos-Reason1 = 7B & 56B with the Fig 14–15 scores	Research page vs arXiv v1	CAVEAT — see below
12	Cosmos 3: mixture-of-transformers omnimodel; Super/Nano/Edge; #1 on open leaderboards	NVIDIA newsroom (Jun 2026)	OK
13	Cosmos 3 data = 20T tokens, ~1B images, ~400M videos	Press coverage (not NVIDIA)	CAVEAT — see below
14	Platform families: Predict / Transfer / Reason + Tokenizer / Curator / RL / Guardrails	Cosmos docs + DeepWiki	OK
15	Predict 1 variants (diffusion 7/14B, AR 4/12B, 5/13B V2W, upsampler, decoder)	arXiv 2501.03575 + GitHub	OK
16	Transfer = (Multi-)ControlNet on Predict; controls depth/edge/seg/LiDAR/HD-map	Cosmos docs + GitHub	OK
17	Transfer 2.5 built on Predict 2.5; 2B is 3.5× smaller than Transfer1-7B; control blocks distributed 1 per 7 (vs at start)	NVIDIA research page	OK — exact
18	Reason 1 = hybrid Mamba-MLP-Transformer; 16K context; SFT + RL	arXiv 2503.15558 (html)	OK — exact
19	Reason 2 = 2B/8B; 256K context (up from 16K); OCR + 2D/3D localization; #1 open	NVIDIA/HF Reason 2 blog	OK — exact
20	Cosmos 3 = two-tower MoT (Nano 16B = 8+8, Super 64B = 32+32); ViT/VAE encoders; 3D mRoPE	NVIDIA/HF Cosmos 3 blog	OK
21	Cosmos 3 reasons + generates in ONE unified forward pass; separate weights, joint attention (vs the old 2-model pipeline). Causal/bidirectional masking is the typical pattern, not NVIDIA-confirmed.	NVIDIA/HF + dev blog	OK (masking inferred)

Table 14. Claims audit. Green = verified against a primary source; amber = caveat.

The two caveats, in detail

- **! Cosmos-Reason1 size & scores (claim 11):** NVIDIA's research page describes a released **7B** model with the scores plotted in Figures 14–15. The **arXiv v1** of the paper instead describes an **8B** model and reports different numbers (e.g. physical common sense 52.3% vs 54.3%; intuitive-physics RL gain 65.7→68.7 vs 74.5→81.5). This is a paper-revision difference. This guide uses the **research-page (7B)** figures because that matches the publicly released checkpoint — but treat the exact decimals as version-dependent. The qualitative conclusions (bigger model helps; RL helps) hold

in both.

- **! Cosmos 3 data scale (claim 13):** the “20 trillion tokens / ~1B images / ~400M videos” figures come from **press coverage** of the launch. NVIDIA's own newsroom release only says “billions of samples across text, image, video, sound and action.” Treat the precise counts as **unconfirmed by NVIDIA**.

Bottom line: the guide is accurate where primary sources exist. The only soft spots are (a) decimal-level benchmark numbers for Cosmos-Reason1, which differ between paper versions, and (b) Cosmos 3's exact data counts, which NVIDIA has not officially published.

Putting it all together

The four generations trace a clear arc:

- **Cosmos 1 (Toolbox):** two engines (diffusion + GPT), a strong causal tokenizer, T5 text, 9-category data, fully documented hyperparameters. Maximum flexibility.
- **Cosmos 2 (Polish):** diffusion-only, sparse attention for up to 2.6× speed, action-conditioned post-training. Same idea, executed better.
- **Cosmos 2.5 (Unify):** one flow-based model replaces three; Cosmos-Reason1 reads the prompt; 200M clips; 30s multi-camera video; RL + model-merge post-training.
- **Cosmos 3 (Leap):** mixture-of-transformers omnimodel; reasons before generating; adds audio and actions; ~20T-token training; Super/Nano/Edge tiers; #1 open model on many leaderboards.

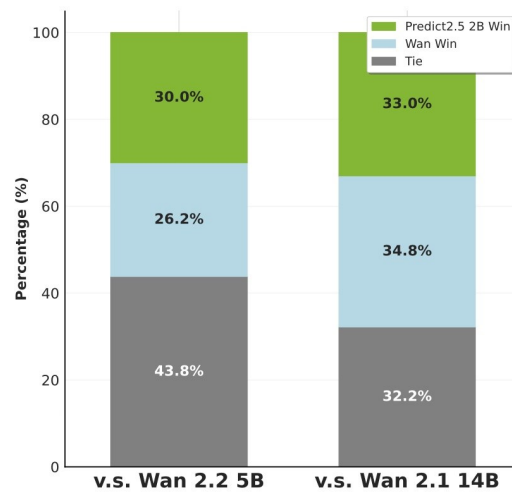
The single sentence to remember

Cosmos went from “**many tools that dream video**” (1) → “**the best tool, polished**” (2) → “**one smart unified tool**” (2.5) → “**a reasoning, seeing, hearing, acting omnimodel**” (3).

Appendix A — Example results (from NVIDIA's pages)

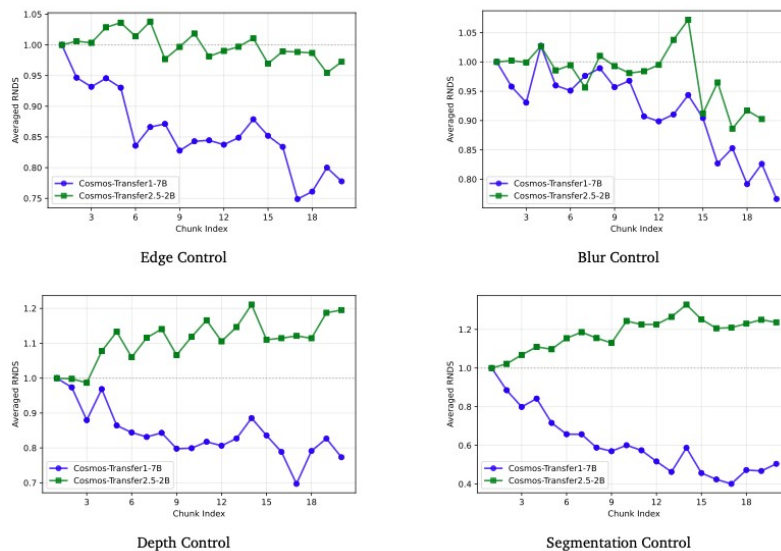
These are real result figures and example inputs taken from NVIDIA's official Cosmos pages, reproduced here for educational reference. © NVIDIA; see the link under each image.

A1 · Cosmos Predict 2.5 — human-preference win rate vs. Wan baselines



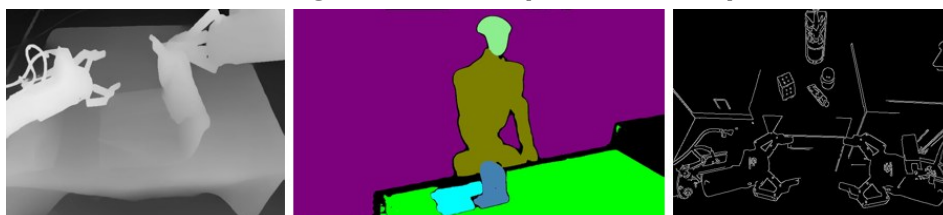
Predict 2.5 (2B) “win / tie / lose” against Wan 2.2 (5B) and Wan 2.1 (14B) in human preference — a small model competitive with much larger baselines. Source: research.nvidia.com/labs/cosmos-lab/cosmos-predict2.5/

A2 · Cosmos Transfer 2.5 (2B) vs. Transfer 1 (7B) — quality across long videos



Averaged RNDs (higher = better) across video “chunk index” for Edge / Blur / Depth / Segmentation control. Transfer 2.5-2B (green) stays high while Transfer 1-7B (blue) degrades over long sequences — the 3.5× smaller model is also better. Source: research.nvidia.com/labs/cosmos-lab/cosmos-transfer2.5/ (Figure 8)

A3 · What Cosmos Transfer ingests — example control inputs



Left→right: depth, segmentation and edge control maps. Cosmos Transfer turns structured inputs like these into photorealistic video (sim-to-real). Source: research.nvidia.com/labs/cosmos-lab/cosmos-transfer2.5/

Appendix B — References & further reading

Primary papers

- **Cosmos World Foundation Model Platform for Physical AI (Cosmos 1)**
<https://arxiv.org/abs/2501.03575>
- **Cosmos-Reason1: From Physical Common Sense to Embodied Reasoning**
<https://arxiv.org/abs/2503.15558>
- **Mixture-of-Transformers: A Sparse, Scalable Multi-Modal Architecture**
<https://arxiv.org/abs/2411.04996>
- **Generalized Neighborhood Attention (sparse attention / NATTEN)**
<https://arxiv.org/abs/2504.16922>

NVIDIA research & model pages

- **Cosmos-Predict2.5**
<https://research.nvidia.com/labs/cosmos-lab/cosmos-predict2.5/>
- **Cosmos-Transfer2.5**
<https://research.nvidia.com/labs/cosmos-lab/cosmos-transfer2.5/>
- **Cosmos 3 technical report**
<https://research.nvidia.com/labs/cosmos-lab/cosmos3/technical-report.pdf>
- **Cosmos model families (docs / DeepWiki)**
<https://deepwiki.com/nvidia-cosmos/cosmos-cookbook/3-cosmos-model-families>

Blogs & announcements

- **HF: Cosmos Predict 2.5 & Transfer 2.5**
<https://huggingface.co/blog/nvidia/cosmos-predict-and-transfer2-5>
- **HF: Cosmos Reason 2 brings advanced reasoning**
<https://huggingface.co/blog/nvidia/nvidia-cosmos-reason-2-brings-advanced-reasoning>
- **HF: Welcome Cosmos 3 for Physical AI**
<https://huggingface.co/blog/nvidia/cosmos-3-for-physical-ai>
- **NVIDIA blog: Develop Physical AI with Cosmos 3**
<https://developer.nvidia.com/blog/develop-physical-ai-reasoning-world-and-action-models-with-nvidia-cosmos-3/>
- **NVIDIA newsroom: Cosmos 3 launch (Jun 2026)**
<https://nvidianews.nvidia.com/news/nvidia-launches-cosmos-3-the-open-frontier-foundation-model-for-physical-ai>

GitHub

- **cosmos-predict1 / predict2 / predict2.5**
<https://github.com/nvidia-cosmos>
- **cosmos-transfer1 / transfer2.5**
<https://github.com/nvidia-cosmos>
- **cosmos-reason1 / reason2**
<https://github.com/nvidia-cosmos>

Images in Appendix A are © NVIDIA, reproduced from the linked pages for non-commercial educational reference. All numeric claims are quoted from the sources above; where sources disagree or NVIDIA has not published a figure, this is flagged on the verification page. Educational summary.